

Object Oriented Analysis and Design

Mini Project Submission – Phase 1

Project Title: **Research and Collaboration System**

Team Number: **Team-15**

Team Members:

1. Salar Hafeez – PES1UG24CS827

2. MALPATEL Revanth Reddy – PES1UG23CS344

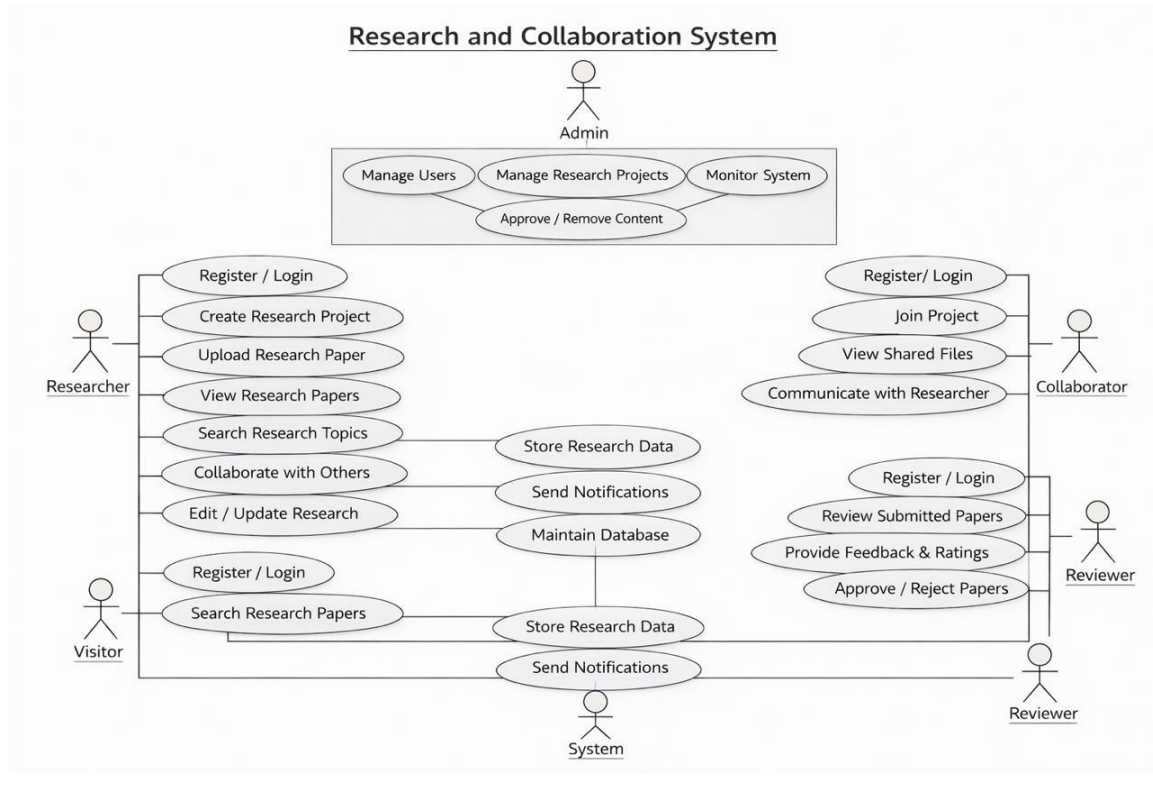
3. Magalur Adarsh – PES1UG23CS338

Phase-1: Use Case Diagram

Brief Description of the System:

- The **Research and Collaboration System** is a software platform developed to manage and support research-related activities efficiently.
- It enables **researchers** to create research projects, upload papers, update research work, and collaborate with others.
- **Collaborators** can join projects, contribute research material, and communicate with researchers.
- **Reviewers** evaluate submitted research papers by reviewing content, providing feedback, and approving or rejecting papers.
- **Visitors** are allowed to browse and search publicly available research papers without modifying content.
- **Administrators** manage users, research projects, system activities, and content moderation.
- The system enforces **role-based access control** to ensure secure and authorized interactions.

Use Case Diagram



Actors Identified in the Use Case Diagram

1. Admin

The admin is responsible for the overall management and maintenance of the system. The admin controls users, research projects, and system activities to ensure smooth and secure operation.

2. Researcher

The researcher is the primary contributor who creates, manages, and updates research projects and research papers within the system.

3. Collaborator

The collaborator supports the researcher by joining research projects, contributing content, and communicating during the research process.

4. Reviewer

The reviewer evaluates submitted research papers, provides feedback, and decides whether the papers should be approved or rejected.

5. Visitor

The visitor is a general user who can browse and search publicly available research papers without modifying any content.

Use Case Explanation

Admin

- Manage Users: Controls user accounts and access permissions.
- Manage Research Projects: Oversees research projects in the system.
- Monitor System: Tracks system activities to ensure security and performance.
- Approve or Remove Content: Maintains quality and compliance of content.

Researcher

- Register / Login: Accesses the system securely.
- Create Research Project: Initiates new research projects.
- Upload Research Paper: Submits research papers to the system.
- View Research Papers: Views available research papers.
- Search Research Topics: Searches for relevant research areas.
- Collaborate with Others: Works with collaborators and reviewers.
- Share Files / Documents: Shares research-related files.
- Edit / Update Research: Updates and improves research content.

Collaborator

- Register / Login: Accesses the system.
- Join Project: Participates in research projects.
- Upload Contributions: Submits research inputs.
- View Shared Files: Accesses shared project documents.
- Communicate with Researcher: Coordinates with researchers.

Reviewer

- Register / Login: Accesses the system securely.
- Review Submitted Papers: Evaluates research papers.
- Provide Feedback and Ratings: Gives feedback on papers.

- Approve or Reject Papers: Decides acceptance or rejection.

Visitor

- Search Research Topics: Searches research papers by topic.
 - Browse Research Papers: Views publicly available research papers.
-

Conclusion

The Use Case Diagram clearly represents the interactions between different actors and the Research and Collaboration System. It highlights role-based access control, structured collaboration, and effective management of research projects and papers.
